

16th Feb 2026 | DA6401

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⇒ Revision (Quiz 1)

40 Ques (100 marks)

↳ 21 MCQ

14 numericals

5 small derivat^{ns} (assumpⁿ & final derivat^{ns})

Practice Problems set

1.
$$\begin{cases} \text{OR } w_1 = w_2 = w_3 = 1 \\ x_1, x_2, x_3 \in \{0, 1\} \end{cases} \quad \hat{y} = wx + b \geq 0$$

→ $Q = 1$

2.
$$w = [2, -1, 3]^T$$

$$b = -2$$

$$\eta = 0.5$$

$$x = [1, 2, -1]^T \Rightarrow y = -1$$

$$w = w + y \eta x$$

$$w = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} + (-1)(0.5) \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

-0.5
-1
+0.5

$$= [1.5 \quad -2 \quad 3.5]^T$$

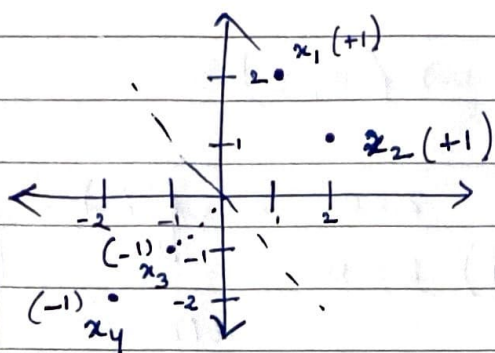
$$\|w_{\text{new}}\| = \sqrt{2.25 + 4 + (3.5)^2}$$

$\hookrightarrow$
 12.25
 $+ 4$
 $\hline 2.25$
 18.50

$$= \sqrt{18.5}$$
$$= 4.3$$

3. B

4.



2 units.

$$\hat{y} = w_1 x_1 + w_2 x_2$$

max. margin $\Rightarrow$ min. distance
 $x+y=0$
 then it would be $\sqrt{2}$

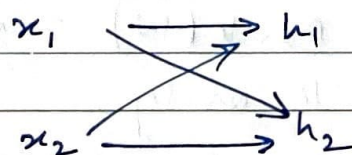
5.

$$W^{(1)} = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

tanh

$$x = [1, 1]^T$$

$$b^{(1)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$



$$\|h\| = 0$$

$$h = (W^T x + b) \tanh$$

$$= \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$2 \times 2 \quad 2 \times 1$

$$= \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

6.

$$\sigma(z), n_{in}, n_{out}, N(0, \sigma_w^2)$$

$$\sigma_w^2?$$

variance of activation is preserved across layers

$$\hat{y} = W^T x \quad \hat{y} = \sum w_{ij} x_j$$

$$\text{Var}(\hat{y}) = \text{Var}\left(\sum w_{ij} x_j\right)$$

$$= n_{in} \sigma_w^2 \sigma_x^2$$

forward pass

$$\sigma^2 w = \frac{1}{n_{in}}$$

$$\text{backward pass} \Rightarrow \sigma^2 w = \frac{1}{n_{out}}$$

$$\sigma_w^2 = \frac{2}{n_{in} + n_{out}}$$

(harmonic mean)

Xavier init

7. finally sigmoid $\Rightarrow (0, 1)$

8. ECU $f(z) = \begin{cases} z & z > 0 \\ \alpha(e^z - 1) & z \leq 0 \end{cases} \quad f'(z) = \begin{cases} 1 & z > 0 \\ e^z & z \leq 0 \end{cases}$

$$f(-1) = 1(e^{-1} - 1)$$

$$= -0.632$$

$$f'(-1) = e^z$$

$$= e^{-1} = 0.368$$

9. $h = \tanh(W^T x + b)$
 $\hookrightarrow W \in \mathbb{R}^{k \times d}$

max. no. of linearly independent vectors in the input space gets preserved.

$$W \in \mathbb{R}^{k \times d}$$

$$\text{Rank}(W) = \min(k, d)$$

at most k only.

10. $y = [0, 1, 0]$

CE

$$e^2 = 7.3$$

logits $\leftarrow z = [2, 1, 0.5]$

$$e^1 = 2.7$$

$$e^{0.5} = 1.649$$

$$\hat{y} = \text{softmax}(z)$$

$$CE = \sum_{i=1}^n y \log(\hat{y}_i)$$

$$\hat{y} = \frac{e^z}{\sum e^z} = [0.629, 0.231, 0.140]$$

$$L = -\log(\text{prob class})$$

$$= -\log(0.231)$$

$$= 1.465$$

11. $f(z) = \max(0.01z, z)$

$\Rightarrow$ Leaky Relu

$$f(-5) = \max(0.05, -5) = -0.05$$

$$f'(z) = \begin{matrix} \max \\ (0.01, 1) \end{matrix} \quad \begin{matrix} z < 0 \\ z > 0 \end{matrix}$$

$$f'(-5) = 0.01$$

12. $W = \begin{bmatrix} 0.5 & -0.3 \\ 0.2 & 0.4 \\ -0.1 & 0.6 \end{bmatrix}$

$\nabla_w L$

$$s = [0.2, -0.4, 0.3]^T$$

$$a_{prev} = [1.5, 0.8]^T$$

$$\boxed{\nabla_w L = s a^T}$$

$$\begin{bmatrix} 0.2 \\ -0.4 \\ 0.3 \end{bmatrix}_{3 \times 1} \begin{bmatrix} 1.5 & 0.8 \end{bmatrix}_{1 \times 2}$$

$$\nabla_w L = \begin{bmatrix} 0.3 & 1.2 \\ -0.6 & -0.32 \\ 0.45 & 0.24 \end{bmatrix}$$

$$\begin{bmatrix} 0.3 & 1.6 \\ -0.6 & -0.32 \\ 0.45 & 0.24 \end{bmatrix}$$

13.

$$a^{L+1} = a^L + F(a^L, w^L)$$

$$\left\| \frac{\partial F}{\partial a} \right\| < 0$$

min. value of $\frac{\partial L}{\partial a^{(i)}}$ relative to $\frac{\partial L}{\partial a^L}$

$$\frac{\partial L}{\partial a^L} = \frac{\partial L}{\partial a^{(L)}} \frac{\partial a^{(L)}}{\partial a^{L-1}} \dots = \frac{\partial a^L}{\partial a^1}$$

for any given layer

$$\frac{\partial a^{(L+1)}}{\partial a^{(L)}} = \left(1 + \frac{\partial F}{\partial a^2} \right) \rightarrow \text{this tends to } 0$$

so, $(1)^L = 1$ (preserves completely)



we want effective batch-size 128 using MB of 32

$$k = \frac{B}{b} = 4$$

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14. $b = 32$, $\sigma_g^2 = 4.0$, $B = 128$

$$\sigma_{MB} = \frac{\sigma^2}{b} = \frac{4}{32} = \frac{1}{8}$$

original $\sigma^2 = 4$ $\frac{4}{32 \times x} = \frac{4}{128}$ $x = \frac{128}{32} \cdot 4$
 $\text{Var} = \frac{4}{4} = 1$

15. BCE $\Rightarrow L = -(y \log \hat{y} + (1-y) \log(1-\hat{y}))$

$2_n = 2_0 \Rightarrow \eta \frac{\partial L}{\partial y} \quad \frac{\partial L}{\partial y} = \hat{y} - y = 0.05$

$\hat{y} = 0.95$

$y = 1$

$\eta = 0.1$

$\Delta Z = Z_{\text{new}} - Z_{\text{old}} = -\eta(\hat{y} - y)$

$\Delta Z = +0.005$

$$a^{(l+1)} = a^l + F(a^l, w^l), \quad \frac{\partial L}{\partial a^{l+1}} = s^{l+1}$$

$\frac{\partial L}{\partial a^l} = ?$

$a^l \Rightarrow a^{l+1}$

$a^l \rightarrow F(a^l, w^l) \rightarrow a^{l+1}$

$\frac{\partial a^{l+1}}{\partial a^l} = 1 + \frac{\partial F}{\partial a^l}$

$$\frac{\partial L}{\partial a^l} = \underbrace{\frac{\partial L}{\partial a^{l+1}}}_{s^{l+1}} \underbrace{\frac{\partial a^{l+1}}{\partial a^l}}_{1 + \frac{\partial F}{\partial a^l}}$$

$$\frac{\partial L}{\partial a^l} = s^{l+1} \left(1 + \frac{\partial F}{\partial a^l} \right)$$

17. $b = 64$, $\eta = 0.01$, $N = 10,000$

(a) how many param. updates?

$$\# \text{ updates} = \left\lceil \frac{N}{b} \right\rceil = \frac{10000}{64} = 157$$

(b) total gradient computaⁿ for 50 epoch

$$\# \text{ grad comp.} = 50 \times 157 = 7850$$

18. $\sigma(z) = \frac{1}{1+e^{-z}} = (1+e^{-z})^{-1}$

$$\frac{d\sigma}{dz} = -1 (1+e^{-z})^{-2} \cdot \frac{d}{dz} (1+e^{-z})$$

$$= -1 (1+e^{-z})^{-2} \cdot e^{-z}$$

$$= \frac{e^{-z}}{(1+e^{-z})^2}$$

————— (1)

$$1 - \sigma(z) = 1 - \frac{1}{1+e^{-z}} = \frac{1+e^{-z} - 1}{1+e^{-z}}$$

$$= \frac{e^{-z}}{1+e^{-z}}$$

$$\sigma(z) (1 - \sigma(z)) = \frac{1}{1+e^{-z}} \cdot \frac{e^{-z}}{1+e^{-z}} = \frac{e^{-z}}{(1+e^{-z})^2} \quad \text{(2)}$$

Hence, (1) & (2) are same.